23 Z transform

What the $\mathscr{L}\text{aplace}$ transform is for continuous time signals, the $\mathbb{Z}$ transform is for discrete time signals. 80% of the content is the same, but there are some key differences.

1) Definition

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Where does this idea come from?

$$z^n \to h[n] \to y[n]$$

We have an eigenvalue property

$$\begin{aligned} y[n] &= z^n * h[n] \\ &= \sum_{k=-\infty}^{\infty} h[k]z^{n-k} \\ &= z^n \underbrace{\sum_{k=-\infty}^{\infty} h[k]z^{-k}}_{H(z)} \end{aligned}$$

2) How to $\mathbb{Z}$ transform?

Plug in to the definition, or use another gosh dang look up table with rules breh

Example:

$$\begin{aligned} x[n] &= a^n u[n] \\ z(z) &= \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} \\ &= \sum_{n=0}^{\infty} (az^{-1})^n \\ &= \frac{1}{1-az^{-1}} \text{ if } \left|az^{-1}\right| < 1 \end{aligned}$$

So

$$a^n u[n] \overset{z}{\leftrightarrow} \frac{1}{1-az^{-1}}$$

Example:

Lets take a system with

$$x[n] = -a^n u[-n-1]$$

$$
\begin{aligned}
X(z) &= \sum_{n=-\infty}^{\infty} -a^{n}u[-n-1]z^{-n} \\
&= -\sum_{n=-\infty}^{-1} a^{n}z^{-n}, && \text{m=-n} \\
&= -\sum_{m=1}^{\infty} a^{-m}z^{m} \\
&= 1-\sum_{m=0}^{\infty}(a^{-1}z)^{m} && \$\$Sowehave \\
&= 1-\frac{1}{1-a^{-1}z} && \text{if } \left|a^{-1}z\right| < 1 \\
&= \frac{1-a^{-1}z}{1-a^{-1}z}-\frac{1}{1-a^{-1}z} \\
&= \frac{-a^{-1}z}{1-a^{-1}z} \\
&= -\frac{1}{az^{-1}-1} \\
&= \frac{1}{1-az^{-1}}
\end{aligned}
$$

\begin{align}
-a^{n}u[-n-1] \stackrel{z}{\leftrightarrow } \frac{1}{1-az^{-1}} & & \left| z \right| <a
\end{align}

This has the same expression, but has a different region of convergence ### Example Lets look at an impulse

\begin{align}
x[n] = \delta[n] \
X(z) = \sum_{n=-\infty}^{\infty} \delta[n]z^{-n}
\end{align}

$Thishasthesiftingproperty, sowehave\$z^{0} = 1\$$

\begin{align}
\delta[n] \stackrel{z}{\leftrightarrow } 1 & & \text{ all z }
\end{align}

Example

\begin{align}
x[n] & = \cos(\omega{0}n)u[n] \
X(z) & = \sum{n=-\infty}^{\infty} \cos(\omega{0}n)u[n]z^{-n} \
& = \sum{n=0}^{\infty} \frac{1}{2}(e^{j\omega{0}n}+ e^{-j\omega{0}n}) z^{-n} \
& = \frac{1}{2} \sum{n=0}^{\infty} (e^{j\omega{0}}z^{-1})^{n} + \frac{1}{2} \sum{n=0}^{\infty} (e^{-j\omega{0}}z^{-1})^{n} \
& = \frac{1}{2} \frac{1}{1-e^{j\omega{0}z^{-1}}} + \frac{1}{2} \frac{1}{1-e^{-j\omega{0}}z^{-1}} & \text{ if } \left| z^{-1} \right| < 1 \
& = \frac{1}{2} \frac{1- e^{-j\omega{0}}z^{-1}+ 1-e^{j\omega{0}}z^{-1}}{(1-e^{j\omega{0}}z^{-1})(1-e^{-j\omega{0}}z^{-1})} \
& = \frac{1}{2} \frac{2- z^{-1} \cdot{2}\cos(\omega{0})}{1- z^{-1}2\cos(\omega{0})+z^{-2}} \
& =\frac{1-\cos(\omega{0})z^{-1}}{1-2\cos(\omega_{0})z^{-1}+z^{-2}}& \left| z \right| >1
\end{align}

Method 2: Table + Properties Note that the region of convergence is *DIFFERENT* then before: If we have a zero at a pole, then

\begin{align}
x[n] = \delta[n] + 3 \cdot 2^{n}u[n] - 2\cos(\omega_{0}n) u[n]
\end{align}

$Letspracticedecomposingasignalwithatable:$

\begin{align}
\delta[n] & \leftrightarrow 1 & \text{ all z } \
2 \cdot 2^{n}u[n] & \leftrightarrow 3 \cdot \frac{1}{1-2z^{-1}} & \left| z \right| > 2 \
-2 \cdot \cos(\omega*{0}n)u[n] & \leftrightarrow -2 \frac{1-\cos(\omega*{0}z^{-1})}{1-2\cos(\omega_{0})z^{-1}+z^{-2}} & \left| z \right| >1
\end{align}

$$Soweget$$

\begin{align}
X(z) = 1+ \frac{3}{1-2z^{-1}} - 2 \frac{1-\cos(\omega*{0})z^{-1}}{1-2\cos(\omega*{0})z^{-1}+ z^{-2}}
\end{align}

$$ThishastheROC\$ |z| > 2\$ becausealltermsmustconverge.$$